CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 37 (2012) 87–93

Contents lists available at SciVerse ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic assessment of Au–Ho and Au–Tm binary systems

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ARTICLE INFO

Article history:
Received 2 November 2011
Received in revised form
7 February 2012
Accepted 8 February 2012
Available online 8 March 2012

Keywords:
Au–Ho
Au–Tm
CALPHAD
Thermodynamic assessment
ab initio calculations

ABSTRACT

Phase relationships in Au–Ho and Au–Tm binary systems have been thermodynamically assessed by using the CALPHAD technique. The existing thermodynamic descriptions of the systems were improved by incorporating the *ab initio* calculated enthalpies of formation of the intermetallic compounds (IMCs) to the assessment. All the binary intermetallic compounds were treated as stoichiometric phases, while the solution phases, including liquid, fcc, and hcp, were treated as substitutional solution phases. Furthermore, the excess Gibbs energies were formulated with the Redlich–Kister polynomial function. As a result, two self-consist thermodynamic data sets for describing the Au–Ho and Au–Tm binary systems have been obtained.

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1. Introduction

Bonding dissimilar materials is one of the essential steps in the fabrication of modern electronic devices. Prominent examples are integrated circuit (IC) and Micro-Electro-Mechanical Systems (MEMS) devices packaging. Typically, the reliability of MEMS devices packaging at the wafer-level depends on the bonding systems to a large extent. There are two primary types of generally used bonding technologies, namely, "dielectric bonding" and "metallic bonding" [1,2]. The latter one has distinguished advantages coming from the metal layer, which can add further functionality to a final 3D IC and MEMS devices, as the metal layer provides not only mechanical but also an electrical and a thermal connection [1,3]. Therefore, it is of utmost importance to obtain a better understanding of the thermodynamics of the related metallic bonding systems. The knowledge about the formation of intermetallic compounds is particularly important.

It is well established that gold, as an element, has many favorable physical and chemical properties (i.e. good thermal and electrical conductivity, excellent corrosion resistance). Consequently, Au and Au metallizations are widely applied as a surface finish with traditional solder materials, as the wire or the pad metal in wire bonding and as the base metal in wafer-level bonding, examples including such systems are Au–Sn and Au–In [4–7].

Rare earth (RE) metals, which can react with almost all the elements in the periodic table, have increasingly been used in MEMS packaging in order to promote bonding [8–10]. For this purpose, RE metals can be used as an alloying elements in Au–Sn-based and Sn–Ag-based solders. This, however, requires that there is sufficient solubility of the RE elements to the solder base metals used. In some cases RE materials are used as one of the bonding layers between two wafers, two chips or wafer/chip, in order to provide a hermetic, conductive bond with small dimensions [11].

Hence, to utilize the full technological potential of the RE elements, it is essential to have a fundamental knowledge about the Au–RE systems, such as information about the phase equilibrium and the related thermodynamic data. Unfortunately, the phase diagrams for Au–RE have not been systematically investigated, with the notable exceptions of the thermodynamic assessment of the Au–X_1 (X_1 = Pr, La and Er) [12,13] and the experimentally investigated Au–X_2 systems (X_2 = Tm, Gd, Ho, Er and Tm) [14,15]. Both Ho and Tm exhibit solid state solubility to Au (fcc) phase (about 5 at%) thus making them potential candidates as alloying elements in Au-based solders. Since there also exists an experimentally measured phase diagram with Au for both of the elements [14], they were chosen for closer investigation. Thus, the thermodynamic assessment of the Au–Ho and the Au–Tm binary systems are carried out in this work by combining *ab initio* techniques [16,17] with the CALPHAD approach [18]. Two self-consistent thermodynamic data sets were obtained for the Au–Ho and the Au–Tm systems, respectively. The paper is organized as follows: a brief review of previous work is presented in Section 2, all the adopted methods and models are described in Section 3, the results and discussions can be found in Section 4, and conclusions drawn from the simulation are listed in Section 5.

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doi:10.1016/j.calphad.2012.02.002

Table 1
Enthalpies of formation at 0 K obtained by an *ab initio* calculation, the determined data at 298 K, and the assessed values at 298 K. Reference states: Au (fcc), Ho (hcp) and Tm (hcp).

Phase	Prototype [14]	This work, *ab initio* (kJ/mol)	Wang et al. [36] *ab initio* (kJ/mol)	Fitzner et al. calorimetry* (kJ/mol)	Meschel et al. calorimetry** (kJ/mol)	This work, assessed (kJ/mol)	Lattice parameters (pm) [14]		
							a	*b*	*c*
Ho (fcc)	Fcc	−1.18	−1.18	–***	–***		496.7		
$AuHo_2$	Co_2Si	−64.1	–	–***	–	−65.5	702.5	489.22	884.82
αAuHo	CrB	−87.5	–	−81.8 ± 3.6	–	−86.0	368.5	1083.5	459.5
βAuHo	CsCl	−64.1	–	–	–	−67.3	354.1		
$Au_{10}Ho_7$	Gd_7Au_{10}	–	–	–	–	−83.7	1021		2772
Au_2Ho	$MoSi_2$	−85.4	–	–	−84.6 ± 1.4	−81.3	367.7		894.0
Au_3Ho	$TiCu_3$	−70.8	–	–	−67.1 ± 2.1	−68.1	605.7	496.4	507.2
$Au_{51}Ho_{14}$	$Gd_{14}Ag_{51}$	–	–	–	–	−63.1	1255		913.5
Au_4Ho	$MoNi_4$	−58.5	–	–	–	−59.4	665.3		417.4
Au_6Ho	Au_6Sm	−40.8	–	–	–	−42.9	1029		966
Tm (fcc)	Fcc	−2.68	−2.68	–	–		491.8		
$AuTm_2$	Co_2Si	−65.5	–	–	–	−70.0	695.8	483.4	876.7
αAuTm	CrB	−90	–	–	−92.8 ± 1.9	−90.5	362.0	1078	457.0
βAuTm	CsCl	−87.5	–	–	–	−68.9	351.9		
$Au_{10}Tm_7$	Gd_7Au_{10}	–	–	–	–	−87.6	1008		2748
Au_2Tm	$MoSi_2$	−59.4	–	–	−91.6 ± 2.3	−82.7	364.4		889.6
Au_3Tm	$TiCu_3$	−71.5	–	–	−72.7 ± 2.1	−69.2	601.6	494.3	507.1
Au_4Tm	$MoNi_4$	−59.5	–	–	–	−61.6	663.4		414.1

* High temperature direct synthesis calorimetry at 1473 ± 2 K [24].
** High temperature direct synthesis calorimetry at 1373 ± 2 K [25].
*** Without values.

2. A brief review of previous work

2.1. The Au–Ho binary system

Before 1986, no complete phase diagram for the Au–Ho binary system was available. Only a limited amount of experimentally measured data had been reported, such as one eutectic reaction involving Au (fcc) [19], the solid solubility of Ho in Au [20] and the crystal structure of compounds, except that of $Au_{10}Ho_7$ [21]. However, even though the crystal characteristics of the intermetallic compounds were determined, melting temperatures of the IMCs were not reported other than that of βAuHo (the high temperature phase). In 1986, Gschneidner et al. [22] summarized and evaluated all the earlier discoveries for this binary system. Based on the results, thermodynamic modeling of the Au–Ho system was performed by Gschneidner et al., including the melting temperatures for the relevant IMCs. Their modeling outcome was reported in Refs. [22,23], including the sketch of proposed phase diagram as well as the relevant crystal structure information. On the other hand, the related thermodynamic data was only included in Ref. [23]. In 2002 Saccone et al. [14] experimentally determined the phase relationships of this system by utilizing X-ray powder diffraction (XRD), optical (OM) and scanning electron microscopy (SEM), electron probe microanalysis (EPMA) and differential thermal analysis (DTA). In [14] more information about Au–Ho phases was reported. In addition, the complete phase diagram including one new compound (i.e. $Au_{10}Ho_7$, involved in two new invariant reactions) and with the relevant crystal structure information was published in [14]. There are 8 compounds and 15 invariant equilibria in the Au–Ho binary system reported, as shown in Tables 1 and 2, respectively.

Regarding the thermochemical information, no mixing enthalpies have been determined, and only enthalpies of formation for 3 compounds (i.e. αAuHo, Au_2Ho and Au_3Ho) in the Au–Ho system have been experimentally measured [24,25]. In 1997, the standard enthalpy of formation of αAuHo was determined by Fitzner and Kleppa [24] with the method of high-temperature direct synthesis calorimetry at 1473 K. In 2004 the enthalpies of formation of Au_2Ho and Au_3Ho (as shown in Table 1) were measured by the same group by means of high temperature direct synthesis calorimetry at 1373 K. Furthermore, Wu et al. [26] have theoretically calculated the enthalpies of formation in this system except for that of $Au_{10}Ho_7$ phase by using modified analytical embedded atom method (EAM). However, their result showed that the value of standard enthalpy of formation of βAuHo (existing at high temperature for the allotropic transformation) was higher than that of αAuHo (existing at low temperature). This phenomenon contradicts the thermodynamic stability theory, since the enthalpy of formation of the low temperature phase in an allotropic transformation should always be more negative than that of the high temperature phase. Therefore, the results from Ref. [26] are not taken into consideration in the present work. Instead, the *ab initio* approach is employed to recalculate the enthalpies of formation for the compounds in Au–Ho binary system.

2.2. The Au–Tm binary system

Similar to Au–Ho system, the entire phase diagram for Au–Tm system had not been established before 1986, either experimentally or by thermodynamic calculations. Partial data on the system however existed. Rider et al. [27] experimentally determined the solid solubility of Tm in Au (fcc) and the temperature of the eutectic reaction involved. The melting temperature of βAuTm was reported by McMasters et al. [21]. In addition, the crystal structures of five stoichiometric intermetallic compounds in this system have been reported [21]. However, the melting temperatures of the IMCs have not been determined, except that of the βAuTm phase. Gschneidner et al. [22] carried out the thermodynamic assessment of the Au–Tm system based on the available information in 1986. The evaluated literature data and the obtained thermodynamic parameters were reported in Refs. [22,23] (in the latter more details were reported). Later, in 2002, just as for the Au–Ho system, Saccone et al. [14] experimentally re-established the phase relationships in this system by means of XRD, SEM, EPMA and DTA. One new phase and two related invariant reactions were reported in Ref. [14]. Nevertheless the allotropic transition temperature of βAuTm $\Leftrightarrow$ αAuTm was still not determined. In the present work, such a transformation temperature is proposed to be 90 K below the melting point of βAuTm, which refers to the same temperature in Au–Ho system due to the common features of the

Table 2
Invariant reactions in the Au–Ho binary system.

Reaction	Composition (Au at%)			T/K	Ref.
$L \Leftrightarrow$ Ho (hcp) + $AuHo_2$	22.0	–**	<1	1258*	[14]
	19.8	–	0.8	1248	This work
$L + \beta$AuHo $\Leftrightarrow$ $AuHo_2$	≈30	–	–	1368*	[14]
	29.3	–	–	1378	This work
$L \Leftrightarrow \beta$AuHo	50.0	–		1973*	[14]
	–	–		1974	This work
βAuLa $\Leftrightarrow \alpha$AuHo	–	–		1883*	[14]
	–	–		1886	This work
$L + \alpha$AuHo $\Leftrightarrow$ $Au_{10}Ho_7$	≈59.5	–	–	1633*	[14]
	62.3	–	–	1622	This work
$L \Leftrightarrow Au_2Ho + Au_{10}Ho_7$	61.5	–	–	1558*	[14]
	63.9	–	–	1593	This work
$L \Leftrightarrow Au_2Ho$	66.7	–		1623*	[14]
	–	–		1609	This work
$L \Leftrightarrow Au_2Ho + Au_3Ho$	72.0	–	–	1503*	[14]
	73.2	–	–	1510	This work
$L \Leftrightarrow Au_3Ho$	75.0	–		1523*	[14]
	–	–		1519	This work
$L + Au_3Ho \Leftrightarrow Au_{51}Ho_{14}$	≈82.5	–	–	1333*	[14]
	82.8	–	–	1333	This work
$L + Au_{51}Ho_{14} \Leftrightarrow Au_4Ho$	≈85	–	–	1223*	[14]
	85.9	–	–	1224	This work
$L + AuHo_4 \Leftrightarrow Au_6Ho$	≈87			1113*	[14]
	88.4	–	–	1115	This work
$L \Leftrightarrow Au_6Ho$ + Au(fcc)	89	–	≈96	1078*	[14]
	89.9	–	94.5	1074	This work

** Composition of Au in related line compounds.
* The error of such data is ±1%. And these data measured by DTA apparatus at 10 K/min.

Table 3
Invariant reactions in the Au–Tm binary system.

Reaction	Composition (at% Au)			T/K	Ref.
$L \Leftrightarrow$ Tm(hcp) + $AuTm_2$	24.0	–**	<0.1	1308*	[14]
	21.5		0.5	1301	This work
$L + \alpha$AuTm $\Leftrightarrow$ $AuTm_2$	≈31.0	–	–	1433*	[14]
	33.6	–	–	1441	This work
$L \Leftrightarrow \beta$AuTm	50.0	–		1993*	[14]
	–	–		1993	This work
βAuTm $\Leftrightarrow \alpha$AuTm		–		1903*	[14]
		–		1904	This work
$L + \alpha$AuTm $\Leftrightarrow$ $Au_{10}Tm_7$	≈60.0	–	–	1618*	[14]
	61.4	–	–	1626	This work
$L \Leftrightarrow Au_2Tm + Au_{10}Tm_7$	62.0	–	–	1593*	[14]
	61.7	–	–	1622	This work
$L \Leftrightarrow Au_2Tm$	66.7	–		1703*	[14]
	–	–		1690	This work
$L \Leftrightarrow Au_2Tm + Au_3Tm$	73.0	–	–	1543*	[14]
	73.5	–	–	1555	This work
$L \Leftrightarrow Au_3Tm$	75.0	–		1573*	[14]
	–	–		1563	This work
$L + Au_3Tm \Leftrightarrow Au_4Tm$	≈83	–	–	1308*	[14]
	83.4	–	–	1310	This work
$L \Leftrightarrow Au_4Tm$ + Au(fcc)	89.0	–	≈93.5	1113*	[14]
	88.6	–	92.6	1113	This work

* The error of such data is ±1%, and these data were measured by DTA apparatus at 10 K/min.
** Composition of Au in related line compounds.

phase diagrams of the two systems. In fact, as described in [21], the CsCl (β) $\rightarrow$ CrB (α) transformations occurred at about 100 K below the melting point of CsCl-type compounds. All the 7 compounds and 11 invariant reactions in Au–Tm system are shown in Tables 1 and 3.

Concerning the thermochemical information about the Au–Tm system, no mixing enthalpies have been reported, and only enthalpies of formation of 3 compounds have been experimentally measured in 2004, i,e, αAuTm, Au_2Tm and Au_3Tm (as shown in Table 1) [25]. Hence, in this work *ab initio* approach is employed to calculate the enthalpies of formation of all the compounds in Au–Tm system except for $Au_{10}Tm_7$ phase.

Consequently, the thermodynamic assessment of Au–Ho and Au–Tm binary systems is carried out in this work. In comparison to the work of Gschneidner et al. [22,23], more information is taken into consideration in the current assessment, including the universally used lattice stabilities for the pure elements, experimentally determined new compounds (i.e. $Au_{10}Ho_7$ and $Au_{10}Tm_7$) and related invariant reactions as well as the theoretically calculated enthalpies of formation for compounds in both Au–Ho and Au–Tm systems.

3. Calculation methods

3.1. ab initio calculations

The *ab initio* approach is used to calculate the lattice stability of the metastable Tm(fcc) and Ho(fcc). Besides, this approach is also employed to calculate the enthalpies of formation for most intermetallic compounds (at 0 K) in the Au–Ho and Au–Tm systems (except for those of $Au_{10}Ho_7$, $Au_{51}Ho_{14}$ and $Au_{10}Tm_7$ compounds), since there were only limited experimentally determined enthalpies of formation for both of these systems. The approach, as implemented in the highly-efficient Vienna *ab initio* simulation package (VASP) [16,17], is performed by using the scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) method [28,29] within the generalized gradient approximation (GGA). For the GGA exchange–correlation function, Perdew–Bucke–Ernzerhof (PBE) [30,31] is employed. A constant plane-wave energy cutoff of 450 eV is adopted for the A_xB_y compounds. Brillouin-zone integrations are performed using Monkhorst–Pack scheme [32] with a 11 × 11 × 11 grid for Au_2R (R = Ho, Tm), Au_4R, αAuR and βAuR, 7 × 8 × 8 for Au_3R, 4 × 4 × 4 for Au_6R, and 6 × 9 × 8 for AuR_2, as well as the Methfessel–Paxton [33] technique with the smearing parameter of 0.2 eV. The reciprocal space (k-point) meshes are increased in order to achieve convergence with a precision of 5 meV/atom. The total energy converged numerically to less than 1×10^{-6} eV/unit with respect to electronic, ionic and unit cell degrees of freedom where the latter two are relaxed by using Hellman–Feymann forces with a preconditioned conjugated gradient algorithm. After the process of structural optimization, the Hellman–Feymann forces on each ion are less than 0.01 eV/Å. All calculations are performed by using the "high" setting within VASP to avoid wrap-around errors. In addition, spin polarization is employed in all calculations.

The enthalpies of formation of the IMCs are calculated by the following equation

$$\Delta H\left(A_xB_y\right) = E_{total}\left(A_xB_y\right) - xE_{total}(A) - yE_{total}(B) \quad (1)$$

where $E_{total}(A_xB_y)$, $E_{total}(A)$, and $E_{total}(B)$ are the calculated total energies (per atom at T = 0 K) of the IMCs as well as for pure A and B, respectively.

3.2. Thermodynamic models

The lattice stability of the element i (i = Au, Ho and Tm, except for Tm(fcc) and Ho(fcc) which were calculated by the *ab initio* approach.) is referred to the constant enthalpy value of the so-called standard element reference H_i^{SER} at 298.15 K and 1 bar as recommended by SGTE (Scientific Group Thermodata Europe), compiled in the paper by Dinsdale [34] as well as the modification from a free online resource [35]. Different models were employed to describe the solution phases and IMCs of the Au–Ho and Au–Tm binary systems.

3.2.1. Solution phases: liquid, Au (fcc) Ho(hcp) and Tm(hcp)

An ordinary substitutional solution model was applied to depict the liquid, fcc and hcp terminal solutions. The molar Gibbs energy

of a solution phase Φ (Φ = liquid, fcc, hcp) can be represented as a sum of the Gibbs energy for the pure components, the ideal entropy term describing a random mixing of components, and the excess Gibbs energy describing the deviation from ideal behavior, i.e.,

$$G^{\phi} = \sum_i x_i {}^0G_i^{\phi} + RT \sum_i x_i \ln(x_i) + {}^{ex}G^{\phi} \quad (2)$$

where R is the gas constant, T is the temperature in Kelvin, x_i is the molar fraction of component i (i = Au, Ho or Tm), and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i, with the crystal structure of ϕ. Here the excess term is formulated as:

$${}^{ex}G^{\phi} = x_{Au}x_R \sum_{j=0,1,\ldots}^{N} {}^{(j)}L_{Au,R}^{\phi}(x_{Au} - x_R)^j \quad (3)$$

where ${}^{(j)}L_{Au,R}^{\phi}$ is the interaction parameter between Au and R (R = Ho or Tm) elements, which was taken to be temperature dependent as follows:

$${}^{(j)}L^{\phi} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1}. \quad (4)$$

Parameters a and b were optimized in this work, the remaining parameters were set to zero.

3.2.2. Stoichiometric intermetallic phases

No significant solubility of the IMCs has been reported [14] and therefore all the intermetallic phases, a total of 6 in the Au–Ho binary system and 7 in the Au–Tm binary system, were treated as line compounds. The Gibbs energy per mole of formula unit Au_mR_n (R = Ho, Tm) is expressed as:

$$G^{Au_mR_n} = mGHSER_{Au} + nGHSER_R + \Delta G_f^{Au_mR_n} \quad (5)$$

where $\Delta G_f^{Au_mR_n}$ is the Gibbs energy of formation of per mole of formula unit of the Au_mR_n. *GHSER* is the Gibbs energy for the pure elements referred to the enthalpy for its stable state at 298.15 K. Due to lack of experimental measurements, it is assumed that the Neumann–Kopp rule applies to the heat capacity, i.e., $\Delta C_p = 0$. Thus the $\Delta G_f^{Au_mR_n}$ can be expressed as:

$$\Delta G_f^{Au_mR_n} = A + BT. \quad (6)$$

Again, A and B were assessed in the present work.

4. Results and discussion

4.1. ab initio calculations

All the calculated enthalpy values are listed in Table 1 together with the standard enthalpies of formation (at 298 K) obtained from the literature [24,25,36], as well as the assessed values in the present work. The values include the enthalpies of formation of all the intermetallics compounds (at 0 K) in the Au–Ho and Au–Tm binary systems excluding those of $Au_{10}Ho_7$, $Au_{51}Ho_{14}$ and $Au_{10}Tm_7$ compounds and the lattice stability of the metastable Ho(fcc) and Tm(fcc). As shown in Table 1, the same values were obtained for the lattice stability of the metastable Ho(fcc) by comparing the *ab initio* results in this work with that from literature [36], as was the value of Tm(fcc). Comparing the calculated values with experimentally measured data, it can be seen that all the calculated results are in good agreement with the measured ones. The deviations between the values are less than 5%. Further, the calculated data exhibit the same general trend with experimental outcomes (Figs. 1 and 2). The most obvious exception is the minimum standard enthalpy of formation in the Au–Ho system, which is located at 50 at% Au instead of the experimentally determined value of 66.7 at% Au. However, based on the data from other similar Au–RE binary systems [13,25], the maximum in the standard enthalpy of formation of the IMCs in the system should, in fact, be located at the composition of 50 at% Au and not at 66.7 at%. Hence, these theoretically calculated enthalpy values can be considered as acceptable. In the present work, the enthalpies of formation based on *ab initio* calculation are adopted later on as the initial thermodynamic parameters (A in Eq. (6), as described in Section 3.2.2) for the relevant compounds. Figs. 1–2 illustrate all of the data obtained that is the optimized results, the *ab initio* calculated values and the experimental data.

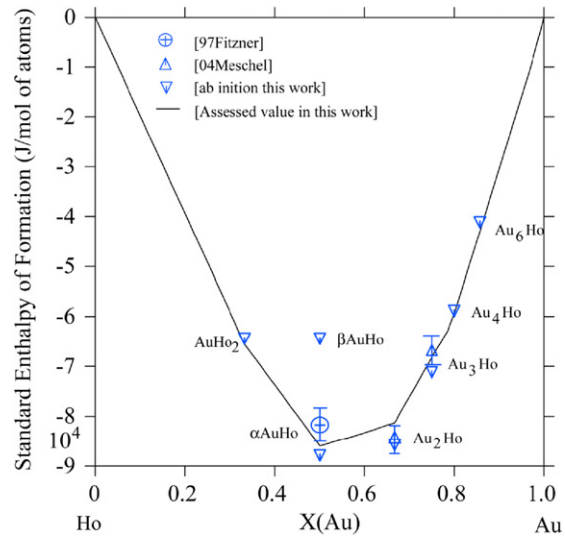

Fig. 1. Standard enthalpy of formation of IMCs in Au–Ho binary systems, reference states: Au (fcc) and Ho(hcp).

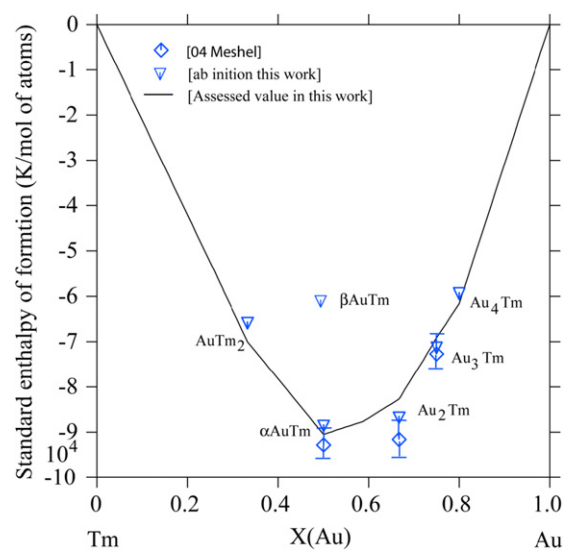

Fig. 2. Standard enthalpy of formation of IMCs in Au–Tm binary systems, reference states: Au (fcc) and Tm (hcp).

4.2. Thermodynamic assessment

Table 4 lists the thermodynamic parameters optimized by using the PARROT module in the Thermo-Calc software [37] based on the experimental and theoretical data available for the Au–Ho and Au–Tm systems. As discussed in the following sections, the present calculations can reproduce most of the experimentally determined and theoretically calculated data.

Table 4
Thermodynamic parameters of the Au–Ho and Au–Tm binary systems obtained in this work.

Phase	Thermodynamic parameters
Liquid model: (Au, Ho)	${}^{0}L^{Liq}_{Au,Ho} = -273007.89 + 10.0045T$
	${}^{1}L^{Liq}_{Au,Ho} = -86622.58$
Fcc model: (Au, Ho)	$G^{Fcc}_{Ho} = 1180 + GHSERHO$
	${}^{1}L^{Fcc}_{Au,Ho} = -362710.98 + 13.0472T$
Hcp model: (Au, Ho)	${}^{0}L^{Hcp}_{Au,Ho} = -126093.45$
$AuHo_2$ model: $Au_{0.333}Ho_{0.667}$	$G^{AuLa_2}_{Au:La} = -65528.56 + 6.5780T + 0.333GHSERAU + 0.667GHSERHO$
αAuHo model: $Au_{0.5}Ho_{0.5}$	$G^{\alpha AuHo}_{Au:Ho} = -85966.80 + 4.1561T + 0.5GHSERAU + 0.5GHSERHO$
βAuHo model: $Au_{0.5}Ho_{0.5}$	$G^{\beta AuHo}_{Au:Ho} = -67341.49 - 5.7206T + 0.5GHSERAU + 0.5GHSERHO$
$Au_{10}Ho_7$ model: $Au_{0.588}Ho_{0.412}$	$G^{Au_{10}Ho_7}_{Au:Ho} = -83700.22 + 4.4206T + 0.588GHSERAU + 0.412GHSERHO$
Au_2Ho model: $Au_{0.667}Ho_{0.333}$	$G^{Au_2Ho}_{Au:Ho} = -81307.48 + 4.9803T + 0.667GHSERAU + 0.333GHSERHO$
Au_3Ho model: $Au_{0.75}Ho_{0.25}$	$G^{Au_3Ho}_{Au:Ho} = -68124.02 + 2.5183T + 0.75GHSERAU + 0.25GHSERHO$
$Au_{51}Ho_{14}$ model: $Au_{0.785}Ho_{0.215}$	$G^{Au_{51}Ho_{14}}_{Au:Ho} = -63056.97 + 3.8214T + 0.785GHSERAU + 0.215GHSERHO$
Au_4Ho model: $Au_{0.8}Ho_{0.2}$	$G^{Au_4Ho}_{Au:Ho} = -59408.63 + 3.6752T + 0.8GHSERAU + 0.2GHSERHO$
Au_6Ho model: $Au_{0.857}Ho_{0.143}$	$G^{Au_6Ho}_{Au:Ho} = -42960.63 + 1.6301T + 0.857GHSERAU + 0.143GHSERHO$
Liquid model: (Au, Tm)	${}^{0}L^{Liq}_{Au,Tm} = -305457.81 + 30.0006T$
	${}^{1}L^{Liq}_{Au,Tm} = -107912.29 + 29.0164T$
	${}^{2}L^{Liq}_{Au,Tm} = +35809.47$
Fcc model: (Au, Tm)	$G^{Fcc}_{Tm} = 2680 + GHSERTM$
	${}^{0}L^{Fcc}_{Au,Tm} = -191775.12 + 2.0549T$
	${}^{1}L^{Fcc}_{Au,Tm} = -119934.20 + 3.9198T$
Hcp model: (Au, Tm)	${}^{0}L^{Hcp}_{Au,Tm} = -125309.51$
$AuTm_2$ model: $Au_{0.333}Tm_{0.667}$	$G^{AuTm_2}_{Au:Tm} = -70003.75 + 8.1953T + 0.333GHSERAU + 0.667GHSERTM$
αAuTm model: $Au_{0.5}Tm_{0.5}$	$G^{\alpha AuTm}_{Au:Tm} = -90475.80 + 7.3693T + 0.5GHSERAU + 0.5GHSERTM$
βAuTm model: $Au_{0.5}Tm_{0.5}$	$G^{\beta AuTm}_{Au:Tm} = -68906.90 - 3.9611T + 0.5GHSERAU + 0.5GHSERTM$
$Au_{10}Tm_7$ model: $Au_{0.588}Tm_{0.412}$	$G^{Au_{10}Tm_7}_{Au:Tm} = -87599.11 + 7.8903T + 0.588GHSERAU + 0.412GHSERTM$
Au_2Tm model: $Au_{0.667}Tm_{0.333}$	$G^{Au_2Tm}_{Au:Tm} = -82702.49 + 7.0140T + 0.667GHSERAU + 0.333GHSERTM$
Au_3Tm model: $Au_{0.75}Tm_{0.25}$	$G^{Au_3Tm}_{Au:Tm} = -69155.84 + 5.2254T + 0.75GHSERAU + 0.25GHSERTM$
Au_4Tm model: $Au_{0.8}Tm_{0.2}$	$G^{Au_4Tm}_{Au:Tm} = -61604.31 + 7.3028T + 0.8GHSERAU + 0.2GHSERTM$

4.2.1. Au–Ho binary system

The thermodynamic assessment of the Au–Ho binary system is carried out in the present work, complemented with the theoretically calculated enthalpies of formation of the relevant IMCs. The compositions and temperatures of the invariable reactions in the binary system are listed in Table 2, including both the experimental data and the assessed values. A comparison between the phase diagram assessed in this work and the data from Ref. [14] was done. As shown by Fig. 3, a good agreement between the two sets of data was obtained, with the exception of the temperatures of the two invariant reactions involving $Au_{10}Ho_7$ and Au_2Ho phases. These are the eutectic reaction ($L \Leftrightarrow Au_2Ho + Au_{10}Ho_7$) and the congruent melting ($L \Leftrightarrow Au_2Ho$). The deviations between the temperatures obtained in the current work and the ones proposed in Ref. [14] (i.e. 1558 and 1623 K) are 35.4 and 13.6 K, respectively. However, the discrepancies can be considered insignificant due to the following facts:

(1) According to Ref. [14], the temperature accuracy of the measured values is ±1%. Therefore, the assessed congruent melting temperature for Au_2Ho (1609 K) is reasonable, compared to the proposed 1623 K.
(2) As discussed in [38], if the curvature of the liquidus of a given compound is very sharp in comparison to those of the neighboring compounds, then it will not be, in general, stable at low temperatures. As there is no experimental evidence to show that the Au_2Ho phase will decompose at low temperatures, the proposed liquidus must be adjusted to be flatter. It means that it is essential to reduce the temperature difference between the related invariant reactions, i.e. the eutectic reaction ($L \Leftrightarrow Au_2Ho + Au_{10}Ho_7$) and the congruent melting ($L \Leftrightarrow Au_2Ho$). Therefore, the temperature of this eutectic reaction should be higher than that of the proposed reaction temperature in Ref. [14].
(3) The proposed invariant temperatures in [14] were based on either the heating curve or cooling curve of DTA analysis, with experimental error of ±1% of measured values. According to the DTA data, the $Au_{10}Ho_7 + Au_2Ho$ dual-phase structure appeared at 1581 K (on heating) and 1556 K (on cooling). The proposed eutectic reaction temperature (1558 K) in Ref. [14] was based on the value obtained during cooling only. It is known that the cooling data are commonly more problematic than the heating data. Therefore, values based on both the heating and the cooling curves, would have been more reliable, as they would minimize the problems related both to superheating and undercooling phenomena in DTA.
(4) In order to fit the already existed experimental enthalpy of formation of Au_2Ho [25], the optimization work could be carried out consistently only by utilizing the temperatures determined in this study.

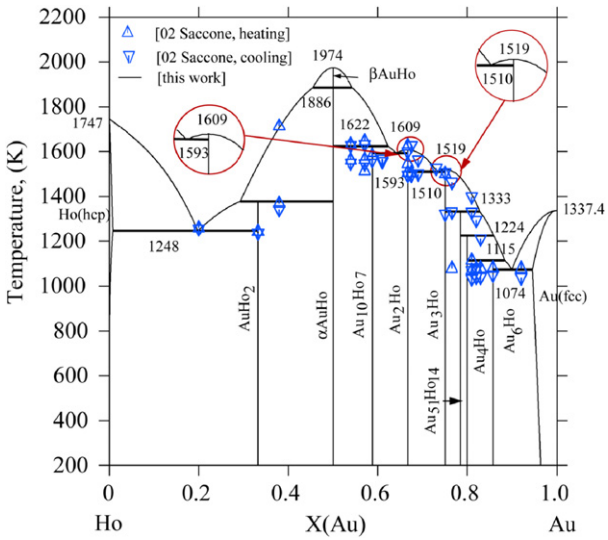

Fig. 3. Calculated phase diagram of Au–Ho binary system, with DTA value from Ref. [14].

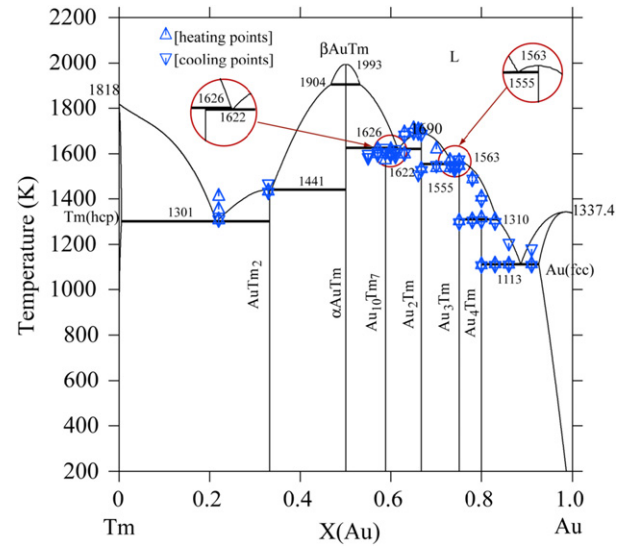

Fig. 4. Calculated phase diagram of Au–Tm binary system, with DTA value from Ref. [14].

With regard to the thermochemical information, the standard enthalpies of the formation of IMCs in the Au–Ho system can also be reproduced by using the obtained thermodynamical parameters. As illustrated in Fig. 1, the optimized results fit well with the *ab initio* approach calculated values and experimentally measured data [24,25].

4.2.2. Au–Tm binary system

The phase diagram of the Au–Tm system shares many of the features of the Au–Ho system. As an exception, in the latter system there are two IMC phases, namely, those with the stoichiometric ratio of 51(Au):14 and 6(Au):1, which do not exist in the Au–Tm system. The assessed phase diagram of the Au–Tm system is shown in Fig. 4, with the data from Ref. [14] superimposed. In addition, Table 4 lists all of the thermodynamically assessed compositions and temperatures of the relevant invariant reactions and the experimental data from Ref. [14]. The assessed results are consistent with the experimental data except for the temperatures of the two invariant reactions involving $Au_{10}Tm_7$ and Au_2Tm compounds. These are the eutectic reaction ($L \Leftrightarrow Au_2Tm + Au_{10}Tm_7$) and the congruent melting ($L \Leftrightarrow Au_2Tm$), in which the optimized eutectic reaction temperature is 28.7 K higher than that proposed in Ref. [14] (1593 K), and the assessed congruent point is 13 K lower than that reported in [14] (1703 K). However, just as discussed in Section 4.2.1, the present results are acceptable due to the error of the experimental values (±1%) [14], the requirement of thermodynamic stability [38], and the experimentally measured standard enthalpies of formation of compounds [25]. In addition, as discussed above, in order to minimize the problems related to superheating and undercooling both the heating and the cooling curves should be utilized when determining the temperatures involving invariant reactions. Instead, the proposed eutectic temperature (1593 K) in [14] is almost the lowest temperature of the related cooling curve (1589 ∼ 1618 K). Hence, as in the case of the Au–Ho system, also here the differences in the temperatures of the two invariant reactions between the current work and that of Ref. [14] can be ignored.

The assessed standard enthalpies of formation of IMCs in this system are illustrated in Fig. 2, together with the experimental data [25] and the *ab initio* results. As can be seen the data agree well with each other.

5. Conclusions

Enthalpies of formation of most of the IMCs in Au–Ho and Au–Tm systems at 0 K have been theoretically calculated by an *ab initio* approach in this paper. The phase relations in the two Au–R (R = Ho and Tm) binary systems have been thermodynamically assessed by utilizing the CALPHAD method based on the experimental as well as the theoretical data. Two self-consistent sets of thermodynamic parameters describing various phases in these systems have been obtained, which can reliably reproduce most of the experimental information. These results provide the necessary and sufficient starting point for the assessment work of the important ternary Au–R–Sn or Au–R–In systems.

Acknowledgments

One of the authors, H.Q. Dong, is thankful for the financial support from the Graduate School in Electronics Telecommunications and Automation, Finland. X.M Tao thanks the National Natural Science Foundation of China (11047031) and the Guangxi Natural Science Foundation (2011GXNSFC018003).

Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2012.02.002.

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